

1. Coding Question: Secure Note-Taking API

Problem Statement:

You are tasked with building a simple ASP.NET Core Web API for a secure note-taking application. The API should allow users to register, log in, and manage their notes securely using **JWT tokens** for authentication.

Requirements:

1. **User Registration:**
 - Create an endpoint for users to register with their username and password.
 - Ensure that the password is securely hashed before being stored in the database.
2. **User Login:**
 - Create an endpoint for users to log in using their username and password.
 - Upon successful login, generate and return a **JWT token** that will be used for authenticating future requests.
3. **Manage Notes:**
 - Create endpoints to add, retrieve, update, and delete notes. Each note should be associated with a user.
 - Ensure that only authenticated users can access these note management endpoints, requiring the JWT token in the authorization header.
4. **Input Validation:**
 - Validate that the username is unique during registration and that both username and password meet basic security requirements.

Sample Input:

1. **Registration Request** to `/api/auth/register`:

```
{
  "username": "notetaker",
  "password": "SecureNote123!"
}
```
2. **Login Request** to `/api/auth/login`:

```
{
  "username": "notetaker",
  "password": "SecureNote123!"
}
```
3. **Add Note Request** to `/api/notes` (with JWT token in Authorization header):

```
{
  "title": "Grocery List",
  "content": "Eggs, Milk, Bread"
}
```

Sample Output: 1.**Registration Response:**

```
{  
  "message": "User registered successfully. Please log in."  
}
```

} 2. Login**Response:**

```
{  
  "token": "<jwt-token>",  
  "expires_in": 3600,  
  "user": {  
    "username": "notetaker"  
  }  
}
```

} 3. Add Note**Response:**

```
{  
  "message": "Note added successfully.",  
  "noteId": 1  
}
```

Explanation:

- **User Registration:** The user registers with a username and password, which is hashed and stored securely. The response indicates successful registration.
- **User Login:** Upon logging in, the user receives a JWT token, which will be used to authenticate requests to manage notes.
- **Note Management:** The user can add notes only after logging in and providing the JWT token, ensuring secure access to their notes.